



Mansoura University
Faculty of Computers and Information
Department of Information Systems
Second Semester- 2023-2024



Database Systems

Lecture 01: What Is a Database System?

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OUTLINE



Course Description

Introduction

ANSI/SPARK Architecture

Database Management System (DBMB)

Roles in Database Environment

History

COURSE DESCRIPTION

- The main objective of this course is to provide students with the background to design, implement, and use database management systems.
 - What is a database system?
 - Database System Concepts and Architecture
 - The Relational Data Model and Relational Database Constraints
 - Data Modeling Using the Entity-Relationship (ER) Model
 - Basics of Functional Dependencies and Normalization for Relational Databases
 - Basic SQL
 - Transaction processing
 - Distributed databases

OUTLINE



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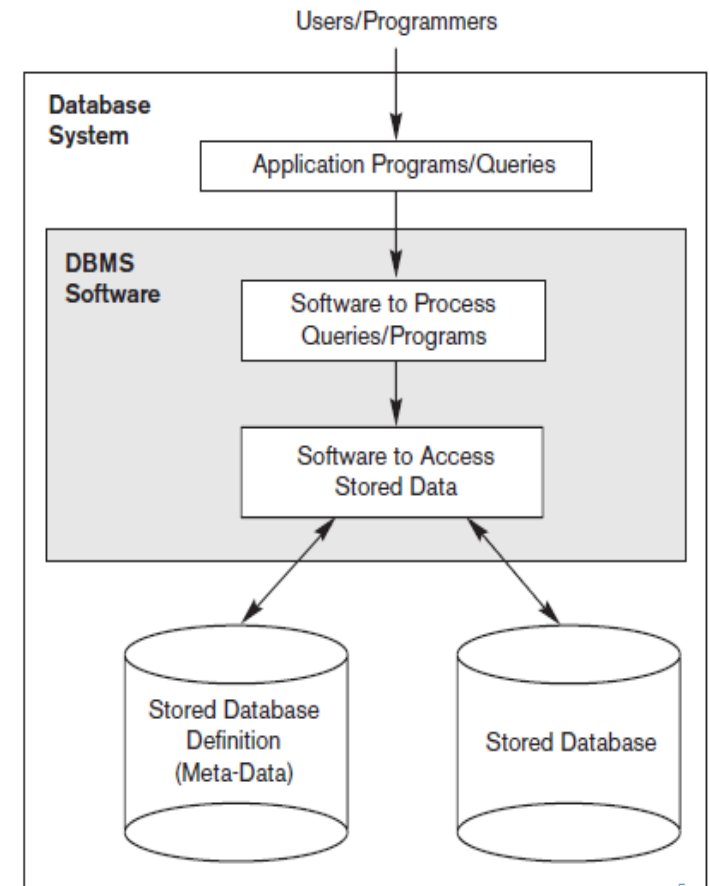
Database Management System (DBMB)

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INTRODUCTION

- A **Database** is a collection of related data.
- The **Database Management System (DBMS)** is the software that manages and controls access to the database.
- A **Database Application** is simply a program that interacts with the database at some point in its execution.
- A **Database System** contains not only the database itself but also a complete definition or description of the database structure and constraints stored in the **catalog** (called **meta-data**). The **database and the metadata are managed by DBMS** and accessed by **database applications**.



DATABASE MODELS

- **Flat Files:**
 - Data simply dumped in file.
 - No structure, no DBMS.
- **Hierarchical Files:**
 - Organize files in tree structure.
 - Parent- Child, each child has only one parent (one-to-many).
- **Network Files:**
 - Multiple inverted tree.
 - Child may have more than one parent.
- **Relational DB:**
 - Data organized in tabular form (Tables)
 - Each table(relation) can be accessed directly
- **Object-oriented DB:**
 - Data represented as objects.

TRADITIONAL FILE-BASED SYSTEMS

- A **file** is simply a **collection of records**, which contains logically related data.
- **Each record** contains a logically connected **set of one or more fields**, where **each field** represents **some characteristic of the real-world object** that is being modeled.

LIMITATIONS OF THE FILE-BASED APPROACH

- **Data redundancy and inconsistency**
 - Multiple file formats, duplication of information in different files.
- **Difficulty in accessing data**
 - Need to write a new program to carry out each new task.
- **Data isolation** — multiple files and formats
- **Integrity problems**
 - Integrity constraints (e.g., account balance > 0) become “buried” in program code rather than being stated explicitly.
 - Hard to add new constraints or change existing ones.

LIMITATIONS OF THE FILE-BASED APPROACH (CONT.)

- **Atomicity of updates**
 - Failures may leave database in an inconsistent state with partial updates carried out.
 - Example: Transfer of funds from one account to another should either complete or not happen at all.
- **Concurrent access by multiple users**
 - Concurrent access needed for performance.
 - Uncontrolled concurrent accesses can lead to inconsistencies.
- **Security problems**
 - Hard to provide user access to some, but not all, data.
- **Database systems offer solutions to all the above problems**

DATABASE APPROACH

- **Database:**
 - A shared collection of logically related data, and a description of this data, designed to meet the information needs of an organization.
 - It is a collection of data stored on your computer that may be as simple as a text file or as complex as a relational database.
- **Database Model:**
 - It reflects **basic structure** which can be characterized by **how they organize the connections between different records** depending on the type of applications using the database.

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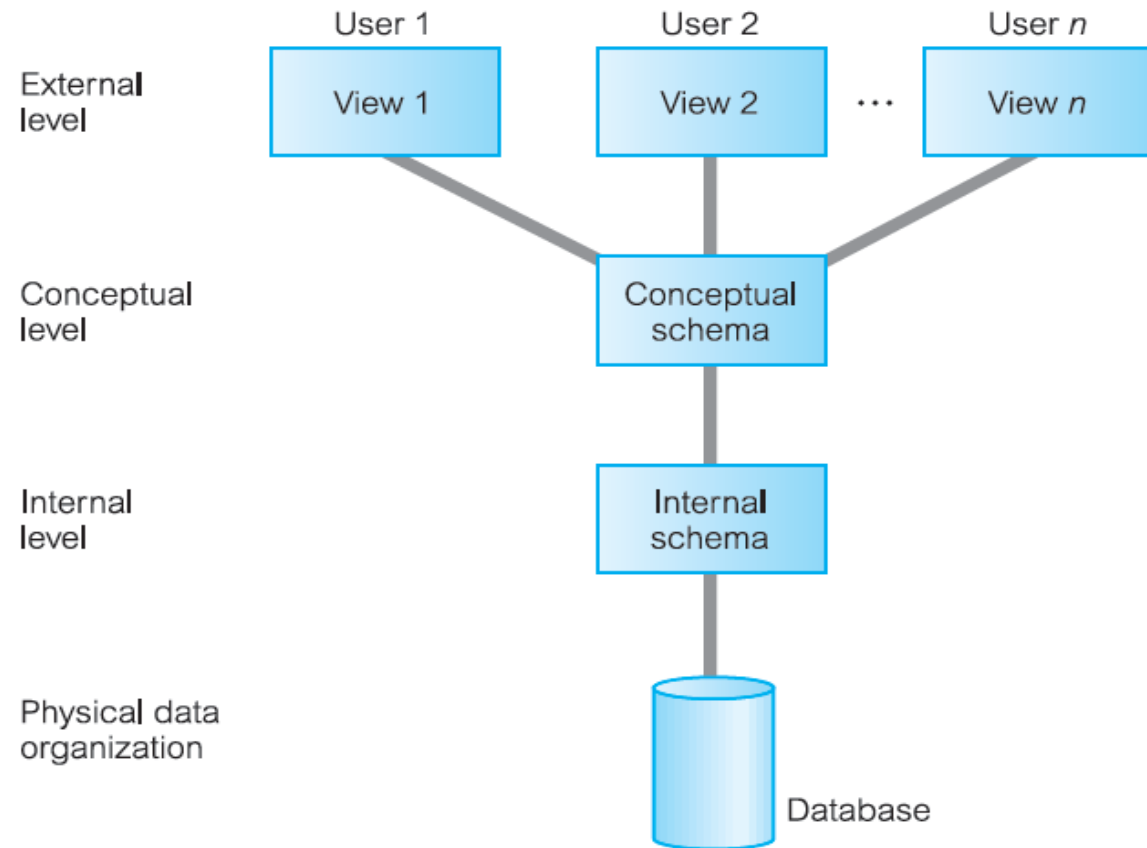
History

ANSI/SPARC

- ANSI
 - American National Standards Institute
- SPARC
 - Standards Planning and Requirements Committee
- 1975
 - Proposed a framework for DBS

THREE-LEVEL ANSI-SPARC ARCHITECTURE

- **External level (Database Users)**
 - Provide a view of the database tailored to a user
 - Data gathering
- **Conceptual level (Database Designer)**
 - Deals with the organization of data as a whole
 - Constraints and security
 - Entity-relationship diagram
 - Normalization
- **Internal level (System Designer)**
 - Deal with physical storage of data
 - Storage, index, compression, & encryption
 - File organization



DATABASE SCHEMAS

External view 1

sNo	fName	lName	age	salary
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External view 2

staffNo	lName	branchNo
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Conceptual level

staffNo	fName	lName	DOB	salary	branchNo
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Internal level

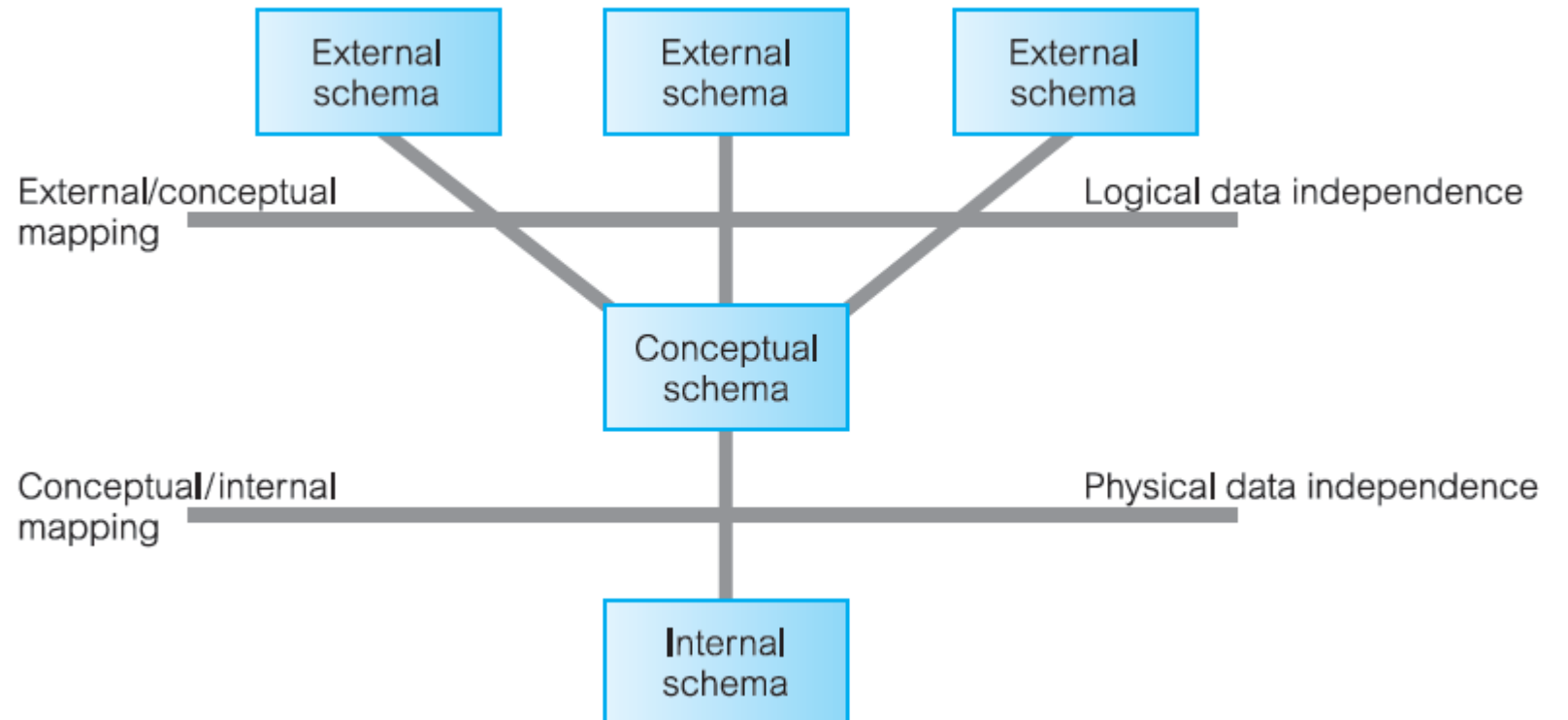
```
struct STAFF {  
    int staffNo;  
    int branchNo;  
    char fName [15];  
    char lName [15];  
    struct date dateOfBirth;  
    float salary;  
    struct STAFF *next;  
};  
index staffNo; index branchNo;
```

/* pointer to next Staff record */

/* define indexes for staff */

MAPPING

- Translate information from one level to the next
 - **External/conceptual mapping**
 - **Conceptual/internal mapping**
- Provide data independence



DATA INDEPENDENCE

- Capacity to change one level of a database system
 - Without having to change the next higher level
- **Logical data independence**
 - Conceptual level changed shouldn't affect external levels
- **Physical data independence**
 - Change to internal level shouldn't affect conceptual level

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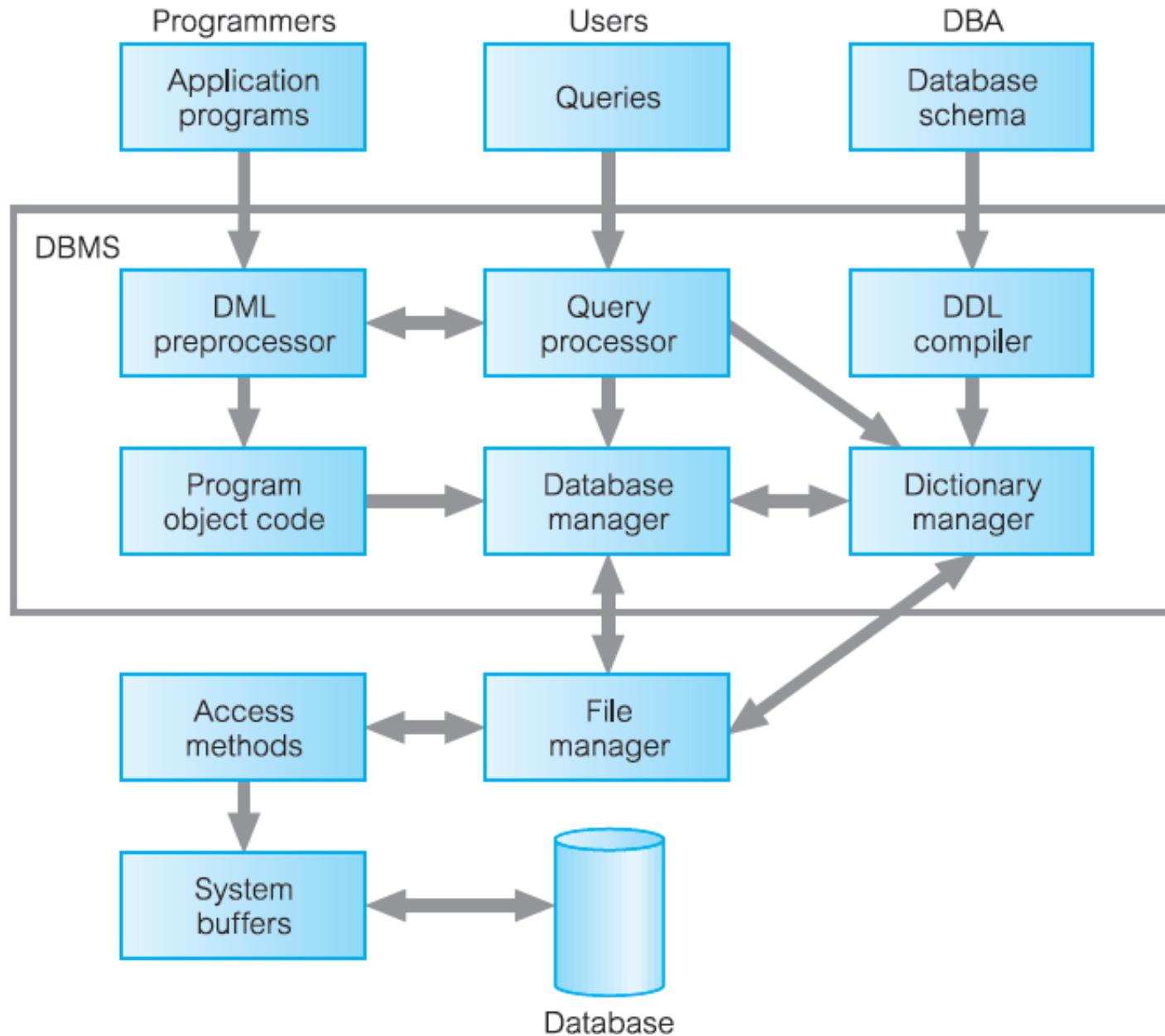
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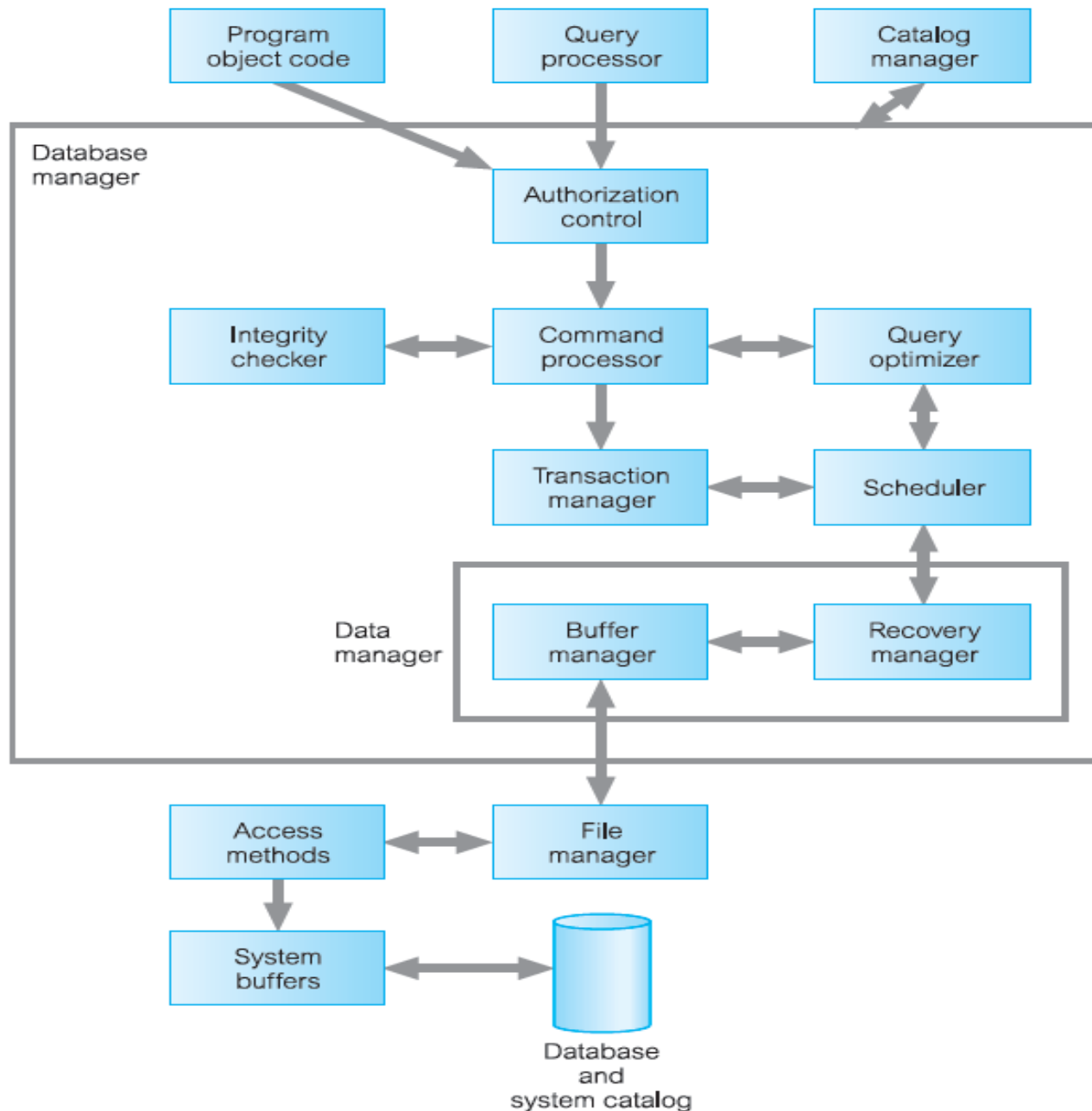
History

THE DATABASE MANAGEMENT SYSTEM (DBMS)

- A DBMS provides the following facilities:
- It allows users to **define the database**, usually through a **Data Definition Language (DDL)**.
- It allows users to **insert, update, delete, and retrieve data** from the database, usually through a **Data Manipulation Language (DML)**.
- It provides **controlled access to the database**. For example:
 - A **security system**, which prevents unauthorized users accessing the database.
 - An **integrity system**, which maintains the consistency of stored data.
 - A **concurrency control system**, which allows shared access of the database.
 - A **recovery control system**, which restores the database to a previous consistent state following a hardware or software failure.
 - A **user-accessible catalog**, which contains descriptions of the data in the database.



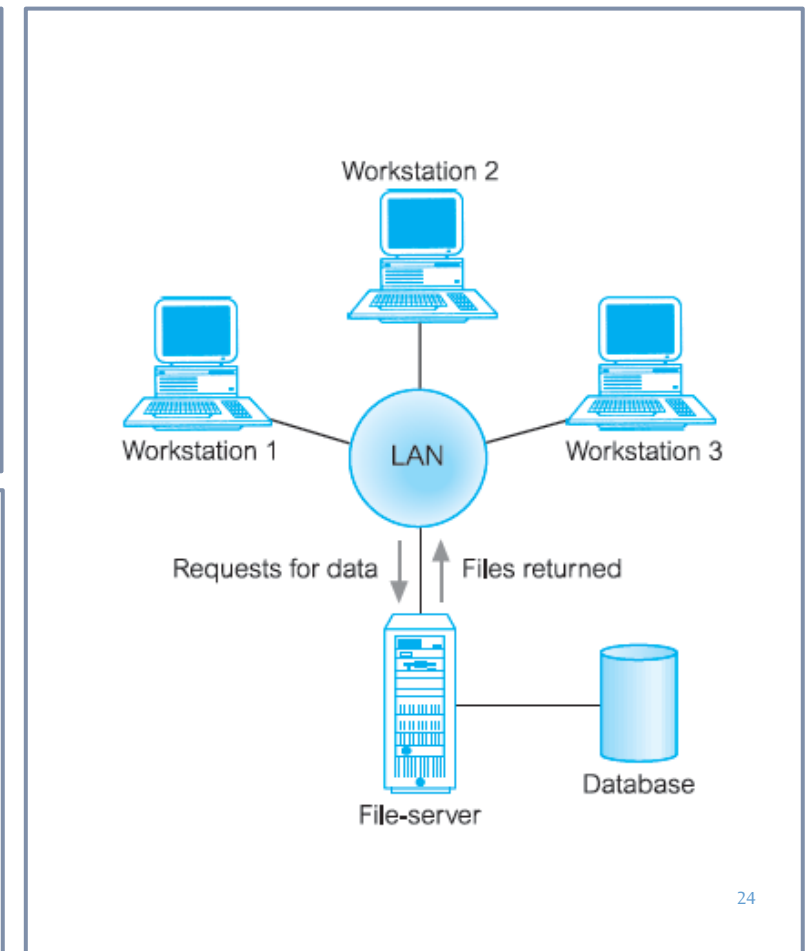
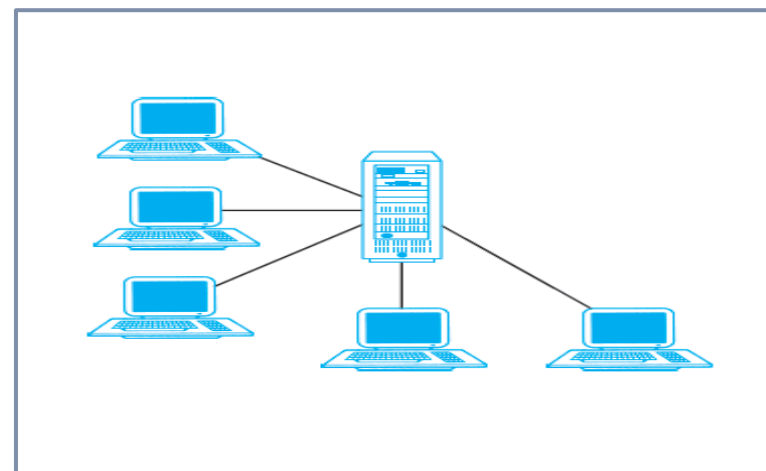
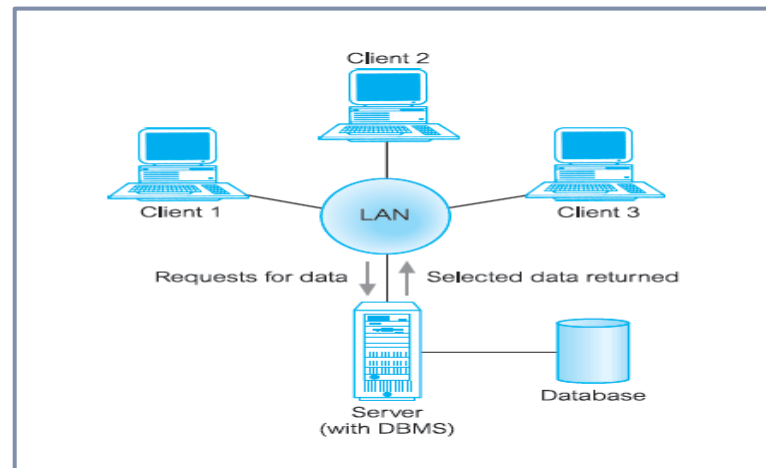
COMPONENTS OF A DBMS



COMPONENTS OF DATABASE MANAGER

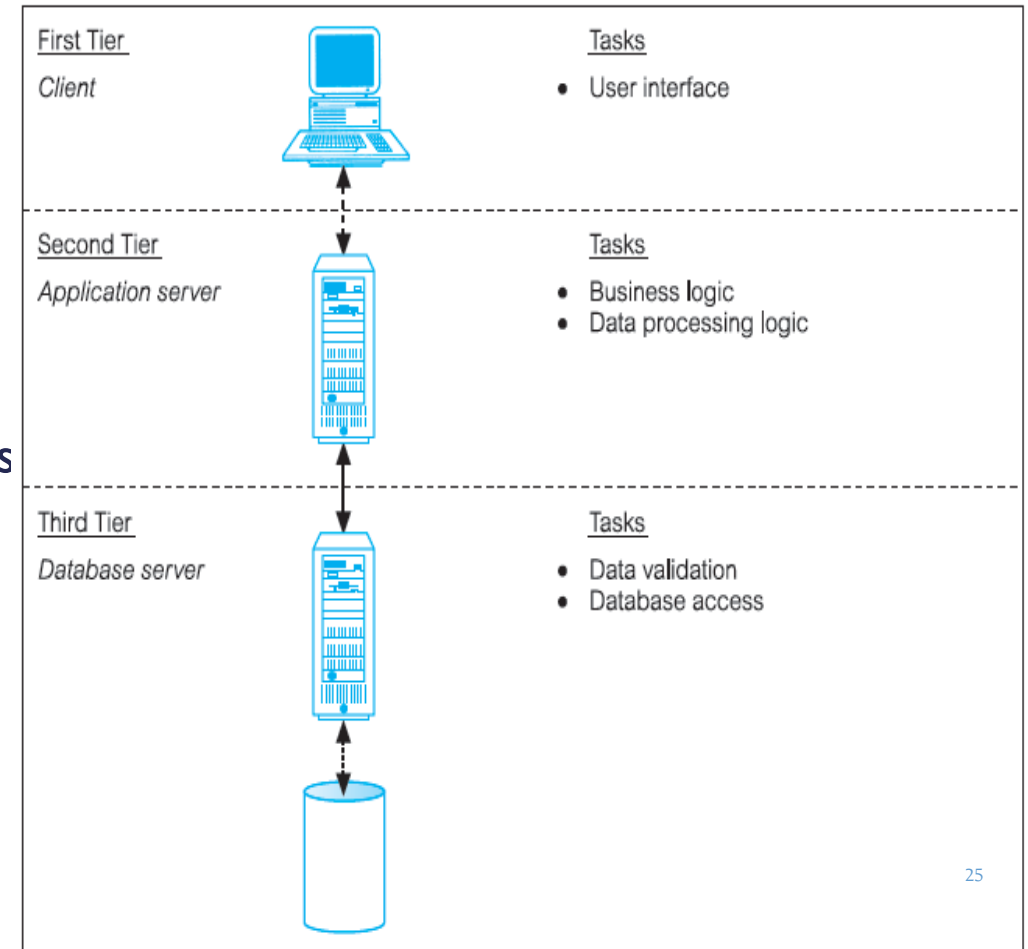
MULTI-USER DBMS ARCHITECTURES

- Teleprocessing
- File server
- Client server

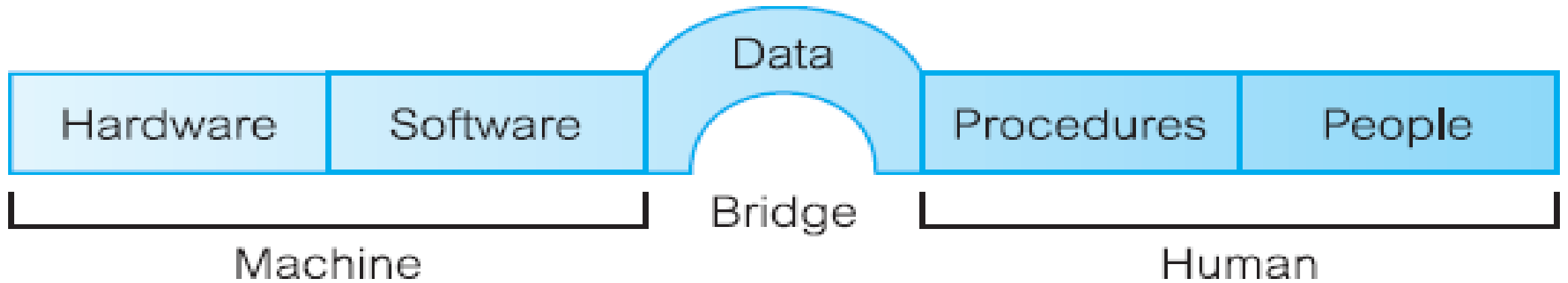


THREE-TIER CLIENT-SERVER ARCHITECTURE

- Three-tier client-server architecture
 - **Client**: user interface
 - **Application server**: business and processing logic
 - **Database server**: data validation and database access
 - **Advantages:**
 - Reducing client cost.
 - Software distribution.
 - Maintenance cost.
 - Balancing load.



COMPONENTS OF THE DBMS ENVIRONMENT



EXAMPLE OF A DATABASE (WITH A CONCEPTUAL DATA MODEL)

- Mini-world for the example:
 - Part of a UNIVERSITY environment.
- Some mini-world entities:
 - STUDENTs
 - COURSEs
 - SECTIONs (of COURSEs)
 - (academic) DEPARTMENTs
 - INSTRUCTORs

EXAMPLE OF A DATABASE (WITH A CONCEPTUAL DATA MODEL)

- Some mini-world relationships:
 - SECTIONs are of specific COURSEs
 - STUDENTs take SECTIONs
 - COURSEs have prerequisite COURSEs
 - INSTRUCTORs teach SECTIONs
 - COURSEs are offered by DEPARTMENTs
 - STUDENTs major in DEPARTMENTs

EXAMPLE OF A SIMPLE DATABASE

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

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ROLES IN THE DATABASE ENVIRONMENT

- **The Data Administrator (DA):**
 - **DA** is responsible for the **management of the data resource** including:
 - **database planning,**
 - **development and maintenance of standards,**
 - **policies and procedures,**
 - The DA consults with and advises **senior managers**, ensuring that the direction of database development will ultimately support corporate objectives.

ROLES IN THE DATABASE ENVIRONMENT (CONT.)

- **The Database Administrator (DBA):**
 - **DBA** is responsible for the **physical realization of the database**, including:
 - Responsible for authorizing access to the database,
 - for coordinating and monitoring its use,
 - Acquiring software and hardware resources, controlling its use
 - and monitoring efficiency of operations
 - The role of the DBA is **more technically oriented** than the role of the DA, requiring detailed knowledge of the target DBMS and the system environment.

ROLES IN THE DATABASE ENVIRONMENT (CONT.)

■ Database Designers:

- The logical database designer is **concerned with identifying:**
 - **the data** (that is, the entities and attributes),
 - the **relationships** between the data,
 - and the **constraints** on the data that is to be stored in the database.
- The physical database designer decides how the **logical database design is to be physically realized**. This involves:
 - **mapping the logical database** design into a set of tables and integrity constraints;
 - **selecting specific storage structures** and access methods for the data to achieve good performance;
 - **designing any security measures** required on the data.

ROLES IN THE DATABASE ENVIRONMENT (CONT.)

■ **Application Developers:**

- Once the database has been implemented, the **application programs that provide the required functionality for the end-users must be implemented**. This is the responsibility of the application developers.

■ **End-Users**

- Naïve users are typically unaware of the DBMS. They **access the database through specially written application programs** that attempt to **make the operations** as simple as possible.
- sophisticated end-user is familiar with the structure of the database and the facilities offered by the DBMS. He/she may use a high-level query language such as SQL to perform the required operations. Some sophisticated end-users may even write application programs for their own use.

ADVANTAGES OF DBMSS

- Control of data redundancy
- Data consistency
- More information from the same amount of data
- Sharing of data
- Improved data integrity
- Improved security
- Enforcement of standards
- Economy of scale
- Balance of conflicting requirements
- Improved data accessibility and responsiveness
- Increased productivity
- Improved maintenance through data independence
- Increased concurrency
- Improved backup and recovery services

DISADVANTAGES OF DBMSS

- Complexity
- Size
- Cost of DBMSs
- Additional hardware costs
- Cost of conversion
- Performance
- Higher impact of a failure

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HISTORY OF DATABASE SYSTEMS

- **Early Database Applications:**

- The Hierarchical and Network Models were introduced in mid 1960s and dominated during the seventies.
- A bulk of the worldwide database processing still occurs using these models, particularly, the hierarchical model.

- **Relational Model based Systems:**

- Relational model was originally introduced in 1970, was heavily researched and experimented within IBM Research and several universities.

- Relational DBMS Products emerged in the early 1980s.

HISTORY OF DATABASE SYSTEMS (CONT.)

- Object-oriented and emerging applications:
 - Object-Oriented Database Management Systems (OODBMSs) were introduced in late 1980s and early 1990s to cater to the need of complex data processing in CAD and other applications.
 - Their use has not taken off much.
- Many relational DBMSs have incorporated object database concepts, leading to a new category called object-relational DBMSs (ORDBMSs)
- Extended relational systems add further capabilities (e.g. for multimedia data, XML, and other data types)

HISTORY OF DATABASE SYSTEMS (CONT.)

- Data on the Web and E-commerce Applications:
 - Web contains data in HTML (Hypertext markup language) with links among pages.
 - This has given rise to a new set of applications and E-commerce is using new standards like XML (eXtended Markup Language).
 - Script programming languages such as PHP and JavaScript allow generation of dynamic Web pages that are partially generated from a database
 - Also allow database updates through Web pages

Thank You